



Department of Computer Applications School of Engineering & Technology

Case study:- 01

Subject: Data Visualization and Story Telling

Program/Branch: BCA

Deadline: 19th November 2025

Code: 13480546

Term/Sec: Vth

Session: 2025-26

Learner's Name:.....Amarjeet.....

Learner's ID:.....

Date handed-in/submitted online:.....

Plagiarism Statement

I declare that, apart from properly referenced quotations, this report is my own work and contains no plagiarism; it has not been submitted previously for any other assessed unit on this or other courses.

Learner's Signature:.....

Date:.....

Learning outcomes and criteria covered by this assignment

The aim of the task is to measure the outcome of students' learning in terms of knowledge acquired, understanding developed and skills or abilities gained in relation to achieve the course outcomes :

Students will be able to:

CO1: Analyze audience needs to select and classify relevant KPIs for dashboard inclusion.

CO2: Illustrate and apply Gestalt principles to organize and structure dashboard layouts effectively.

CO3: Apply design tools to develop interactive, user-centric dashboards that communicate insights.

CO4: Formulate critiques of dashboard designs by analyzing the data-ink ratio and reducing visual clutter.

Specification of Assessment

Students need to create a report and produce the solution for the following:

Q.No	Question	Marks	Bloom's Level
1	Global Trade Flow Visualization		Analyse / (Analysis)

Note:

- **Soft / hard copy not exceeding 700 words. Plagiarism must not be more than 15%. Plagiarism will be checked by faculty. Submit within deadline.**
- **Signed copy to be submitted online within two days of deadline closure. Document found copied from any AI tool or copied from any other student will be marked zero without informing the student. Entire component will be marked zero.**

Global Trade Flow Visualization: A Comprehensive Case Study

Introduction to Global Trade Flow Visualization

Global trade flow visualization is crucial for understanding the multifaceted relationships and economic exchanges between countries. The complexity of global trade data—with numerous countries and volumes flowing in multiple directions—poses challenges for clear interpretation. Traditional geographic maps with trade routes often become cluttered and less informative for large datasets. Therefore, effective visualizations such as chord diagrams, network graphs, and tree maps have been innovated to depict trade origins, destinations, and volumes in intuitive and insightful ways. These techniques enable policymakers, economists, and researchers to analyze trade patterns, economic dependencies, and net trade balances efficiently[1].

The evolution of trade visualization reflects growing demand for data-driven insights in international commerce. As global supply chains become increasingly complex and interconnected, the ability to visualize these relationships has become essential for understanding economic flows, identifying trade dependencies, and supporting strategic decision-making at both national and organizational levels. Global trade reached \$22.6 trillion in 2023, and visualizing this vast volume of transactions across thousands of countries and product categories requires sophisticated analytical tools[2].

Visualization Techniques and Data Insights

Chord Diagrams: A Powerful Visualization Method

One strong method of global trade visualization is the chord diagram. This method represents countries as sectors on a circular plot, each with an assigned color. The trade flow is depicted by ribbons colored according to the origin country. The direction of trade is shown by gaps at the destination sector, and trade volume is proportional to the width of the ribbons and arc size.

For example, a chord diagram depicting Indonesia and Japan's trade showed Indonesia's exports to Japan as a light blue flow and imports from Japan in purple, allowing for rapid assessment of trade balance. This intuitive design enables viewers to quickly identify which countries are major trading partners and whether relationships are balanced or skewed[1]. The visual encoding of color, position, and size allows viewers to process complex trade relationships at a glance.

Key Findings from Trade Data Analysis

In a study using data from 2010 to 2015, chord diagrams revealed critical insights about global trade patterns. The United States emerged as a major trade actor, with imports exceeding exports by up to 55%, reflecting its role as a significant consumer market drawing imports from around the world. Among emerging markets, China stood out as a dominant net exporter, with export volumes growing steadily during the period.

Visualizing these flows helps understand economic roles clearly:

- **Net exporters** drive trade surpluses and demonstrate competitive advantage in manufacturing and production
- **Net importers** often depend on global supply chains and consumer demand dynamics

- **Balanced traders** show sustainable trade relationships with relatively equal exchange volumes

These insights from visualization techniques demonstrate how graphical representation transforms raw trade statistics into actionable economic intelligence[1]. Interactive platforms like the World Bank's WITS (World Integrated Trade Solution) and McKinsey's Global Trade Explorer now enable users to explore bilateral trade flows, product-specific details, and tariff impacts dynamically[3].

Case Study: India's Trade and GDP Impact Analysis

Context and Methodology

A detailed case study focused on India's merchandise trade from 2015 to 2021, visualizing key economic factors affecting GDP growth. The research integrated multiple variables to create a comprehensive picture of trade's impact on economic development:

- Export and import monetary values
- Shares of trade with partner countries
- Export and import growth rates
- Currency exchange rates
- Inflation rates
- GDP growth correlations

Key Findings

By visualizing these multifactor data on trade and economic indicators, the study analyzed how external trade impacts India's GDP growth dynamically. The visualization process highlighted several critical relationships:

Factor	Impact on Trade	Economic Significance
Trade Growth	Positive correlation with GDP	Expansion in exports drives development
Currency Fluctuation	Affects competitiveness	Weak rupee boosts exports
Inflation	Inflates costs	Higher inflation reduces competitiveness
Global Integration	Trade dependency	Supply chain participation increases growth

Table 1: Key factors influencing India's trade and GDP growth (2015-2021)

Trade Growth Impact: The data revealed that periods of accelerated export growth coincided with stronger GDP growth, suggesting positive correlation between international trade expansion and domestic economic development[4].

Currency Fluctuations: Exchange rate volatility showed measurable impact on trade competitiveness. Stronger rupee periods correlated with import growth, while weaker rupee periods

boosted export volumes, demonstrating real-time market dynamics.

Inflation and Trade: Inflation rates influenced both the cost of imports and export competitiveness, showing how macroeconomic factors interconnect with international trade flows.

Strategic Implications

The India case study demonstrates the importance of balanced trade growth and currency strength in supporting GDP rise. Fluctuations in these factors and their correlation illustrated how international trade influences national economic stability and development. This approach demonstrates the importance of interactive visualizations for combining complex economic variables alongside trade data to enhance decision-making for policymakers, central banks, and business strategists[4].

Challenges and Future Directions

Current Limitations

While current visualization methods like chord diagrams offer clarity, they face several limitations:

- **Data Completeness:** Not all nations consistently report trade data, resulting in gaps and incomplete pictures of global trade flows
- **Real-time Updates:** Most visualizations rely on historical data with significant reporting lags
- **Granularity:** While product-level details are available, standard visualizations often show only bilateral country-level flows
- **Interactivity:** Static visualizations limit exploration capabilities for researchers and analysts

Emerging Solutions and Platforms

Future work could enhance global trade visualization through advanced interactivity, time series integration, real-time data incorporation, and multi-dimensional analysis[5]. Platforms such as the World Bank's WITS and McKinsey's Global Trade Explorer provide examples of broader visualization capabilities, including tariff impact analysis and product-level trade flows. Additionally, UN Comtrade and UNCTAD statistics databases now offer weekly updated trade nowcasts using AI-powered methodologies to provide real-time indications of global trade trends[2].

Conclusion

Global trade flow visualization represents a transformative approach to understanding international commerce and economic relationships. By converting complex numerical trade data into intuitive visual representations, these tools empower decision-makers to identify patterns, understand dependencies, and forecast economic trends.

The case studies presented—from Indonesia-Japan bilateral trade to India's comprehensive trade-GDP analysis—demonstrate how visualization techniques reveal insights that raw data alone cannot provide. As global trade continues to evolve with technological disruption, new trade agreements, and changing geopolitical dynamics, sophisticated visualization methods will remain essential for comprehending and managing the interconnected world economy[1][5].

The future of trade visualization lies in combining multiple data sources, enhancing interactivity,

and providing real-time insights that support both strategic policy decisions and business operations in an increasingly complex global marketplace.

References

- [1] NYC Data Science. (2022). Data Visualizing Global International Trade. Retrieved from <https://nycdatascience.com/blog/student-works/data-visualizing-global-international-trade/>
- [2] UNCTAD. (2025). Statistics and data | UN Trade and Development. Retrieved from <https://unctad.org/statistics>
- [3] World Bank. (2024). Trade Data Visualization - WITS. Retrieved from <https://wits.worldbank.org/trade-visualization.aspx>
- [4] Publications EAI. (2024). Visualization Process of International Trade and its impact on economic growth. Retrieved from <https://publications.eai.eu/index.php/sis/article/view/5296>
- [5] Our World in Data. (2025). Growth of global trade. Retrieved from <https://ourworldindata.org/grapher/growth-of-global-trade>